

Publishing and Sharing Reports in Power BI

Easy Level

Q1. Explain why publishing and sharing reports is considered more important than just building dashboards in Power BI Desktop.

Answer: Publishing and sharing reports in Power BI is more important than just building dashboards in Desktop because reports provide deep analysis, context, and drill-down capabilities, while publishing ensures secure distribution, collaboration, and governance across teams. Dashboards alone give only a snapshot; without publishing, insights remain siloed and inaccessible.

Why Publishing & Sharing Matters More

1. Collaboration & Accessibility

- >> Reports published to the Power BI Service can be shared with colleagues, embedded in apps, and accessed via mobile.
- >> Dashboards built in Desktop remain local and isolated, limiting their impact.

2. Governance & Security

- >> Publishing allows organizations to manage permissions, roles, and access control.
- >> Shared dashboards without proper publishing often lead to confusion, scattered links, and security risks.

3. Depth of Analysis vs. Snapshot Monitoring

- >> Reports answer “Why are we performing this way?” with filters, drill-downs, and multi-page analysis.
- >> Dashboards answer “How are we doing right now?” with 5–7 KPIs on one screen.

4. Scalability

- >> Publishing enables apps and workspaces, separating builders from consumers.
- >> This ensures a polished, stable experience for end-users while developers continue iterating in the background.

Q2. Define Power BI Service and explain its role in comparison to Power BI Desktop in the reporting lifecycle.

Answer: The **Power BI Service** is Microsoft’s **cloud-based platform** where reports and dashboards are published, shared, and collaboratively managed. It acts as the **distribution and collaboration layer** in the Power BI ecosystem, enabling secure access, governance, and real-time insights across an organization.

1. Purpose:

>> **Power BI Desktop:** A local development environment used to build reports, connect to data sources, and design visuals.

>> **Power BI Service:** A cloud platform for publishing, sharing, and collaboration across teams.

2. Key Activities:

>> **Desktop:** Connect to data, clean and transform with Power Query, build models and visuals, and save .pbix files.

>> **Service:** Publish reports, share dashboards, manage permissions, and schedule automatic refreshes.

3. Audience:

>> **Desktop:** Primarily for data analysts and report creators.

>> **Service:** Designed for managers, executives, and end-users who consume insights.

4. Accessibility:

>> **Desktop:** Installed on a Windows PC, works offline.

>> **Service:** Accessible via browser or mobile apps, enabling anytime, anywhere access.

5. Collaboration:

>> **Desktop:** Limited collaboration since files are stored locally.

>> **Service:** High collaboration through workspaces, apps, and role-based access.

Q3. What does publishing a report mean in Power BI? List any two outcomes that occur after a report is published.

Answer: Publishing a report in Power BI means moving a report created in Power BI Desktop into the Power BI Service (cloud platform). This process uploads both the report and its underlying dataset so that they can be accessed, shared, and managed centrally. It transforms a locally stored **.pbix** file into a collaborative, governed resource that can be consumed by others across the organization.

Two Key Outcomes After Publishing

1. Report + Dataset Availability in Power BI Service

>> The report and its dataset are uploaded together.

>> The dataset becomes available for scheduled refreshes, reuse, and building additional reports or dashboards.

2. Collaboration & Sharing

- >> Reports can be shared securely with colleagues via workspaces, apps, or dashboards.
- >> Role-based access ensures the right people see the right data, enabling enterprise-wide collaboration.

Medium Level

Q4. List and explain the steps involved in publishing a report from Power BI Desktop to Power BI Service in the correct sequence.

Answer: Here's the **correct sequence of steps** you follow when publishing a report from **Power BI Desktop** to the **Power BI Service**.

01 - Open Power BI Desktop

Start with the report you have created locally.

- >> Ensure your report is complete and saved as a .pbix file
- >> Verify visuals, data model, and transformations are working correctly

02 Sign in to Power BI Service

You must be logged in to publish reports.

- >> Click Sign In at the top-right corner of Power BI Desktop
- >> Enter your organizational or Microsoft account credentials

03 Click Publish

This action initiates the upload process.

- >> Go to the Home ribbon
- >> Select Publish → choose My Workspace or another workspace

04 Choose Destination Workspace

Select where the report should reside in the Service.

- >> Pick My Workspace for personal use
- >> Choose a shared workspace for team collaboration

05 Verify Upload Success

Confirm that the report and dataset are now in the Service.

- >> A confirmation message will appear in Desktop
- >> The report and dataset are visible in Power BI Service under the chosen workspace

06 Share and Collaborate

Enable others to access and interact with the report.

- >> Use Share to invite colleagues
- >> Configure role-based access permissions
- >> Optionally pin visuals to dashboards for monitoring

Q5. Differentiate between Report and Dashboard in Power BI based on structure and usage.

Answer: Here's a clear **heading-wise differentiation** between a **Report** and a **Dashboard** in Power BI based on **structure** and **usage**:

Report – Structure

1. Multi-page document created in Power BI Desktop.
2. Contains detailed visuals, charts, tables, and slicers.
3. Built on a single dataset, but can include multiple pages for deep analysis.
4. Allows drill-down, drill-through, and interactive filtering.

Dashboard – Structure

1. Single-page canvas created in the Power BI Service.
2. Made of tiles pinned from reports (can pull visuals from multiple datasets).
3. Summarizes KPIs and metrics at a glance.
4. Designed for quick monitoring rather than detailed exploration.

Report – Usage

1. Used by analysts and data professionals to explore data in depth.
2. Answers “Why are things happening?” through detailed analysis.
3. Best for operational teams who need context, trends, and granular insights.

Dashboard – Usage

1. Used by executives and decision-makers for high-level monitoring.
2. Answers “How are we doing right now?” with key metrics.
3. Best for tracking performance indicators and receiving alerts.

Q6. Explain the difference between the following three sharing methods in Power BI Service:

- >> Direct Report Sharing
- >> Workspace Access
- >> Power BI Apps

Also mention one use case for each.

Answer: The **three sharing methods in Power BI Service** with their differences and practical use cases:

1. Direct Report Sharing

>> **What it is:** You share a specific report directly with individuals by entering their email addresses.

>> **How it works:** The recipient gets access to that report only (not the entire workspace). Permissions can be set as view or edit.

>> **Use Case:** A data analyst shares a monthly sales report directly with the sales manager for review, without exposing other reports in the workspace.

2. Workspace Access

>> **What it is:** You add users to a workspace where multiple reports, datasets, and dashboards are stored.

>> **How it works:** Members can be assigned roles (Viewer, Contributor, Member, Admin) that define their level of access and control.

>> **Use Case:** A project team working on customer analytics is given workspace access so they can collaborate, edit reports, and manage datasets together.

3. Power BI Apps

>> **What it is:** A packaged collection of dashboards and reports published from a workspace into an app.

>> **How it works:** End-users install the app to access curated content. It provides a clean, user-friendly experience without exposing the underlying workspace structure.

>> **Use Case:** The HR department publishes an app containing employee performance dashboards for executives, ensuring they see only polished, finalized content.

Q7. Describe the four workspace roles available in Power BI and explain how access control is managed using these roles.

Answer: **Four workspace roles in Power BI Service** and how access control is managed through them:

1. Admin:

>> **Permissions:** Full control over the workspace. Can add/remove users, assign roles, publish, edit, delete content, and manage settings.

>> **Access Control Role:** Admins define who gets access and at what level, ensuring governance and compliance.

>> **Example:** The BI manager acts as Admin to control workspace membership and security.

2. Member:

>> **Permissions:** Can create, edit, and publish reports and dashboards. Can also share content with others.

>> **Access Control Role:** Members collaborate actively but don't control workspace settings or user roles.

>> **Example:** A data analyst is made a Member to build and update reports for the team.

3. Contributor:

>> **Permissions:** Can edit existing reports, dashboards, and datasets but cannot manage workspace settings or permissions.

>> **Access Control Role:** Contributors ensure content stays updated but rely on Admins/Members for publishing and sharing.

>> **Example:** A junior analyst is given Contributor rights to update visuals in reports without altering workspace governance.

4. Viewer:

>> **Permissions:** Read-only access. Can view reports and dashboards but cannot edit or publish.

>> **Access Control Role:** Viewers are the end-users who consume insights without changing content.

>> **Example:** Executives are assigned Viewer roles to monitor KPIs without risking accidental edits.

Hard Level:

Q8. Why are Power BI Apps considered more secure and scalable than direct sharing? Explain this from a governance and enterprise usage perspective.

Answer:

From a Governance Perspective:

>> **Centralized Control:** Apps allow administrators to manage permissions at the workspace level, ensuring only curated, approved content is distributed.

>> **Reduced Risk:** Direct sharing often leads to ad-hoc access, scattered links, and potential compliance issues. Apps prevent this by offering a structured, governed distribution method.

>> **Version Consistency:** Updates made in the workspace automatically flow into the app, so users always see the latest version without confusion from multiple shared links.

From an Enterprise Usage Perspective:

>> **Scalability:** Apps are designed for large audiences. Instead of sharing reports one-by-one, you publish a single app that hundreds or thousands of users can install.

>> **User Experience:** Apps provide a clean, organized interface with dashboards and reports grouped logically, making it easier for executives and employees to consume insights.

>> **Role Separation:** Developers work in the workspace, while end-users consume content via the app. This separation ensures stability for consumers while allowing ongoing development behind the scenes.

Q9. Explain Row-Level Security (RLS) in Power BI. How does RLS ensure that different users see different data using the same report?

Answer: Row-Level Security (RLS) is a feature in Power BI that restricts data access at the **row level** based on filters you define. It ensures that different users see only the data that is relevant to them, even when they are all using the **same report**.

How RLS Works

>> **Role Creation:** In Power BI Desktop, you define roles (e.g., Region Manager, Sales Rep) and apply filters to datasets.

>> **Rule Application:** These filters determine which rows of data are visible to users assigned to that role.

>> **User Assignment:** After publishing to the Power BI Service, admins assign users or groups to roles.

>> **Dynamic Filtering:** When a user opens the report, Power BI automatically applies the filter rules, showing only the permitted subset of data.

Why RLS Ensures Different Views

>> **Single Report, Multiple Perspectives:** Instead of creating separate reports for each user, RLS applies filters dynamically.

>> **Security & Compliance:** Sensitive data is hidden from unauthorized users, ensuring privacy and regulatory compliance.

>> **Efficiency:** Reduces duplication of reports and simplifies maintenance, since one report can serve many audiences securely.

Q10. Assume an organization shares reports without using Apps, workspace roles, or RLS. Discuss any three risks or issues that may arise related to security, governance, or data misuse, and explain how Power BI best practices help prevent them.

Answer: Structured explanation of the risks when an organization shares reports **without using Apps, workspace roles, or Row-Level Security (RLS)** — and how Power BI best practices prevent them:

1. Uncontrolled Access & Security Breach

>> **Issue:** Direct sharing without RLS or workspace roles means all users may see the entire dataset, including sensitive information (e.g., salaries, confidential sales data).

>> **Impact:** Leads to compliance violations, data leaks, and misuse of confidential information.

>> **Best Practice Fix:** Implement Row-Level Security (RLS) to restrict data visibility per user role, ensuring each person only sees what they are authorized to view.

2. Governance Breakdown & Version Confusion

>> **Issue:** Sharing reports via ad-hoc links creates multiple versions floating around. Users may act on outdated or inconsistent data.

>> **Impact:** Poor decision-making, lack of trust in BI reports, and audit challenges.

>> **Best Practice Fix:** Use Power BI Apps for structured, governed distribution. Apps ensure everyone sees the latest curated version, reducing confusion and maintaining compliance.

3. Collaboration Chaos & Lack of Role Separation

>> **Issue:** Without workspace roles, every user may have edit rights or no clear boundaries. Reports can be accidentally modified or deleted.

>> **Impact:** Loss of data integrity, accidental overwrites, and reduced accountability.

>> **Best Practice Fix:** Assign workspace roles (Admin, Member, Contributor, Viewer) to enforce clear responsibilities. This ensures developers build, admins govern, and end-users consume insights safely.